

Six-state modulation method for improving the accuracy of fiber optic gyroscopes by reducing electrical crosstalk

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Reducing the influence of electrical crosstalk is crucial for the development of fiber optic gyroscopes. We present a six-state modulation method that takes advantage of highly flexible modulation voltages to eliminate the correlation between the modulation voltage with a fixed phase delay and the demodulation sequence. Theoretical analysis indicates that our approach can eliminate the impact of crosstalk with a fixed phase delay at any modulation depth. Experimental measurements demonstrate that compared to the conventional four-state modulation method, the zero bias is reduced by about $0.3319^\circ/\text{h}$ at the modulation depth of $\pi/2$ when our method is applied. Additionally, at the depths of $\pi/2$, $2\pi/3$, and $7\pi/8$, the bias instability is reduced by approximately 30%, and the angular random walk (ARW) is reduced by over 60%. The results confirm its effectiveness in decreasing electrical crosstalk, which holds profound implications for enhancing the accuracy of fiber optic gyroscopes. © 2024 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

Interferometric fiber optic gyroscopes (IFOGs), one of the most significant fiber optic sensing technology instruments, possess high precision, a long lifetime, an all-solid-state structure, and a broad dynamic range, etc. They are utilized extensively in the land, marine, aerospace, and other areas [1–3]. So improving the accuracy of IFOG has been one of the most sought-after goals. Phase modulation plays a vital role in increasing the accuracy of IFOG. However, in practice, a portion of the modulation voltage's energy is coupled to the photodetector output, thus influencing the demodulation result [1,4,5]. We call this electrical crosstalk. It will increase both the zero bias and the bias instability of the IFOG [6,7], a significant factor in reducing IFOG accuracy.

Square wave biasing modulation and four-state modulation are two commonly used modulation methods. However, both methods have their limitations in the pursuit of high accuracy. For the former, square wave biasing modulation can only demodulate the 2π reset error before or after the ramp reset [8]. At low speeds, the reset period is long, so the correction of the error is not timely. In addition, there is crosstalk under square wave biasing modulation. For the latter, the 2π reset error can be demodulated according to the output signal at each

modulation period [8–10], which ensures timely error correction. Nevertheless, it is also unable to avoid electrical crosstalk because of the high correlation between the modulation voltage and the demodulation sequence [11].

In order to mitigate the effect of electrical crosstalk, various methods have been proposed. Mark *et al.* [12] employed orthogonal sequences to minimize the effect of crosstalk. In 1997, an overmodulation technique was invented to overcome crosstalk [13]. Zhang *et al.* [11] also proposed a six-state modulation method, which employs states different from those in this study and can only eliminate electrical crosstalk at a specific modulation depth of $2\pi/3$.

In this paper, we propose a new modulation method aimed at suppressing electrical crosstalk and thus improving the accuracy of IFOGs. This method also includes six states in each modulation period, but unlike Zhang's six-state modulation [11], it eliminates the effect of crosstalk with a fixed phase delay on the demodulation results at any modulation depth in theory. Our six-state modulation scheme, employing highly flexible modulation voltages, removes the correlation between the modulation voltage and the demodulation sequence. Ultimately, the experimental result is described, strongly supporting the theoretical analysis.

2. BASIC PRINCIPLE AND CONVENTIONAL MODULATION METHODS

A. Principle of IFOG

The Sagnac effect is the theoretical basis for fiber optic gyroscopes. When a fiber optic coil rotates, a phase difference between a pair of counter-propagating coherent light waves on a complete trip is generated as [1,14]

$$\Delta\varphi_s = \frac{2\pi L D}{\lambda c} \Omega, \quad (1)$$

where $\Delta\varphi_s$ is Sagnac phase shift, L is the fiber coil length, D is the coil average diameter, λ is the average light source wavelength, and c is the speed of light in vacuum.

In the simplified IFOG ring optical path shown in Fig. 1, one beam of light signal propagates in the direction of $B \rightarrow \text{PM} \rightarrow \text{fibercoil} \rightarrow A$, and the other beam propagates in the direction of $A \rightarrow \text{fibercoil} \rightarrow \text{PM} \rightarrow B$. The intensity of the interference light detected by the photodetector can be expressed as [15,16]

$$I_D = I_0 [1 + \cos(\Delta\varphi_s + \Delta\varphi_b + \Delta\varphi_{fb})], \quad (2)$$

where I_0 is the intensity of the light signal from the light source, $\Delta\varphi_s$ is the Sagnac phase shift induced by rotation, $\Delta\varphi_{fb}$ is the feedback phase shift to compensate for the Sagnac shift $\Delta\varphi_s$, and $\Delta\varphi_b$ is the bias modulation phase to avoid sensitivity approaching 0 when $\Delta\varphi_s + \Delta\varphi_{fb} \approx 0$. $\Delta\varphi_b + \Delta\varphi_{fb}$ is the extra phase difference applied by the Y-waveguide phase modulator. The light intensity I_D , which carries the rotational rate information, is then converted into an electrical signal, which will be digitally analyzed and processed to extract the rotational rate Ω [3].

To achieve phase modulation, the phase modulator based on LiNbO_3 adjusts the refractive index of the optical waveguide by applying differential voltage signals to produce a modulating electric field at the phase modulation electrodes [3]. Under the voltage V_{PM} , it is capable of applying a phase increment of $K_f V_{PM}$ to the optical signal traveling through the phase modulator [15]. K_f is the modulation coefficient. In the subsequent discussion, the modulating voltage that produces a phase increment of magnitude β is specified as V_β .

B. Square Wave Biasing Modulation

Assume that the IFOG is modulated by a square wave. Under square wave modulation without feedback, $\Delta\varphi_b$ is generally

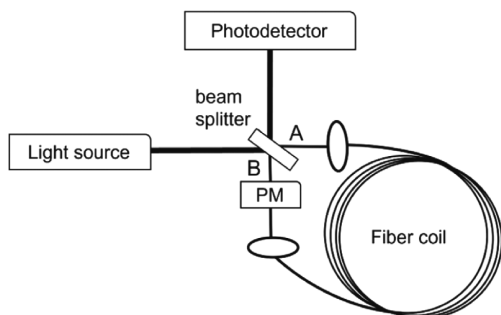


Fig. 1. Simplified IFOG ring optical path. PM, phase modulator.

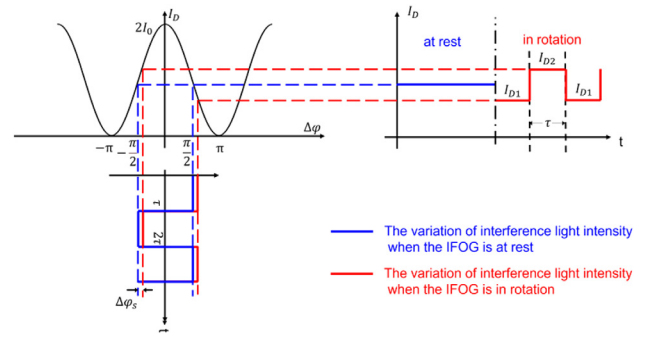


Fig. 2. Variation of I_D when the square wave biasing modulation method is applied.

$\pm\pi/2$, and each state lasts for τ (τ is the transit time of the optical fiber ring). The variation of I_D with time is shown in Fig. 2,

$$\begin{aligned} I_{D1} &= I_0 [1 + \cos(\Delta\varphi_s + \pi/2)] \\ &= I_0 [1 - \sin \Delta\varphi_s], \end{aligned} \quad (3)$$

$$\begin{aligned} I_{D2} &= I_0 [1 + \cos(\Delta\varphi_s - \pi/2)] \\ &= I_0 [1 + \sin \Delta\varphi_s]. \end{aligned} \quad (4)$$

With the sequence shown in Fig. 3 synchronously demodulating I_D , we obtain the demodulation result,

$$\begin{aligned} I_D &= 1 \times I_{D1} + (-1) \times I_{D2} \\ &= -2I_0 \sin \Delta\varphi_s. \end{aligned} \quad (5)$$

The stability of the scale factor, the ratio of the demodulation result to Ω , is an important indicator of the performance of the IFOG. It depends to some extent on the linearity of the IFOG demodulation result. Combining Eqs. (1) and (5), when the IFOG rotates at high speed, the relationship $\sin \Delta\varphi_s \approx \Delta\varphi_s$ is no longer valid, so the linear relationship between the demodulation result and Ω is deteriorated. In order to improve the stability of the scale factor, we introduce negative feedback in the phase modulation link by superimposing a step-wave voltage [Fig. 4(a)] on top of the original modulation voltage [4,15,17]. Due to the fact that $\Delta\varphi_s$ is characterized by low-frequency variations and the time consumed by each demodulation is extremely small, it can be approximated that $\Delta\varphi_{fb}(k) = -\Delta\varphi_s(k-1) \approx -\Delta\varphi_s(k)$. Therefore, $\sin(\Delta\varphi_s(k) + \Delta\varphi_{fb}(k))$ can be approximated as $\Delta\varphi_s(k) - \Delta\varphi_s(k-1)$. By accumulating the output of each demodulation, we can obtain a result with good linearity, as shown in Eq. (6),

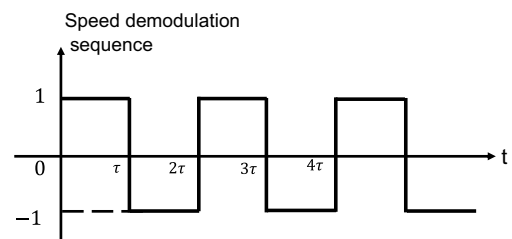


Fig. 3. Speed demodulation sequence for square wave biasing modulation.

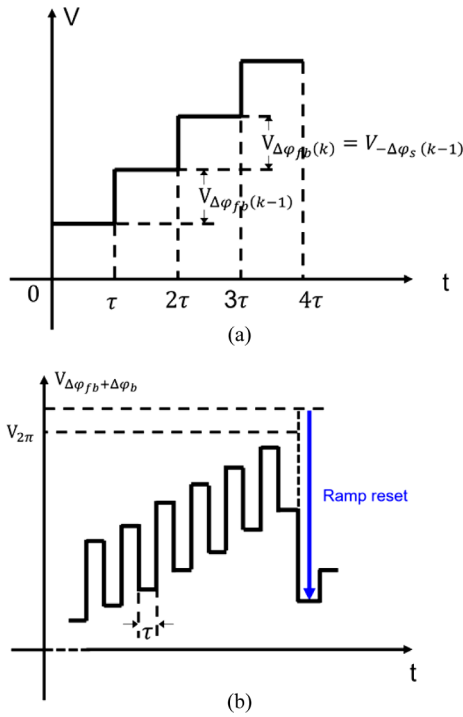


Fig. 4. (a) Stepped waveform voltage for applying feedback phase difference. (b) 2π ramp reset with stepped waveform voltage, where $V_{2\pi}$ is twice the half-wave voltage of the Y-waveguide phase modulator.

$$\begin{aligned}\Sigma_{i=1}^k \text{demod}(i) &= \Sigma_{i=1}^k -2I_0 \sin [\Delta\varphi_s(i) - \Delta\varphi_s(i-1)] \\ &\approx \Sigma_{i=1}^k -2I_0 [\Delta\varphi_s(i) - \Delta\varphi_s(i-1)] \\ &\approx -2I_0 \Delta\varphi_s(k).\end{aligned}\quad (6)$$

However, square wave modulation does have a significant flaw. It performs poorly at low speeds. The parameter of the phase modulator is sensitive to changes in ambient temperature [18,19], thus introducing errors induced by temperature fluctuation in the modulation link. At low rotating speeds, the Sagnac phase difference $\Delta\varphi_s$ is small and the step wave rises slowly, requiring a long time for a 2π ramp reset [Fig. 4(b)]. During the process, error accumulates. The square wave biasing modulation method can only detect the 2π reset error before or after ramp reset [8]. Therefore, it couldn't manage to correct the error in time. To summarize, under square-wave modulation, the IFOG is strongly influenced by the temperature fluctuation of the operation environment at low rotating speeds.

C. Four-State Modulation

Four-state modulation is a commonly used modulation method. It is able to correct the 2π reset error in every modulation period, making up for the shortcoming of square-wave biasing modulation. The four-state modulation approach applies phase difference $\Delta\varphi_b$ to the optical signals according to Fig. 5(a), and the associated modulation voltage is depicted in Fig. 5(b). When feedback is not taken into account, the corresponding interference light intensity I_D is expressed as follows and is further illustrated in Fig. 6:

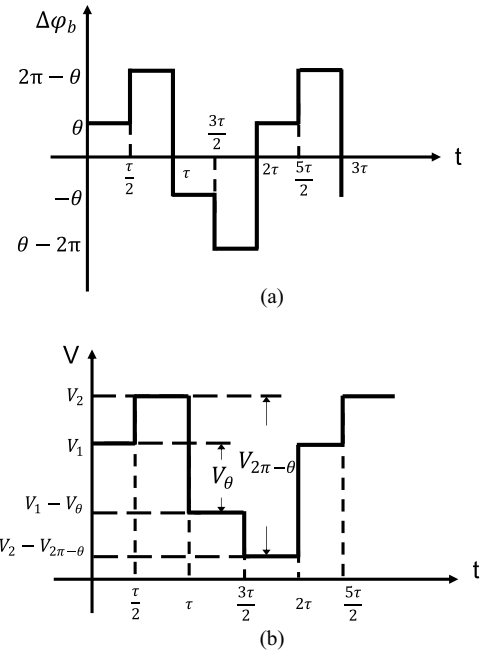


Fig. 5. (a) Phase difference $\Delta\varphi_b$ of the four-state modulation method. (b) The corresponding modulation voltage of the four-state modulation method.

$$\begin{aligned}I_{D1} &= I_0 [1 + \cos(\Delta\varphi_s + \theta)] \\ &= I_0 [1 + \cos \Delta\varphi_s \cos \theta - \sin \Delta\varphi_s \sin \theta],\end{aligned}\quad (7)$$

$$\begin{aligned}I_{D2} &= I_0 [1 + \cos(\Delta\varphi_s + 2\pi - \theta)] \\ &= I_0 [1 + \cos \Delta\varphi_s \cos \theta + \sin \Delta\varphi_s \sin \theta],\end{aligned}\quad (8)$$

$$\begin{aligned}I_{D3} &= I_0 [1 + \cos(\Delta\varphi_s - \theta)] \\ &= I_0 [1 + \cos \Delta\varphi_s \cos \theta + \sin \Delta\varphi_s \sin \theta],\end{aligned}\quad (9)$$

$$\begin{aligned}I_{D4} &= I_0 [1 + \cos(\Delta\varphi_s + \theta - 2\pi)] \\ &= I_0 [1 + \cos \Delta\varphi_s \cos \theta - \sin \Delta\varphi_s \sin \theta].\end{aligned}\quad (10)$$

I_D is synchronously demodulated by the sequence $\{-1, 1, 1, -1\}$ (each state lasts $\tau/2$), and the expression is as follows:

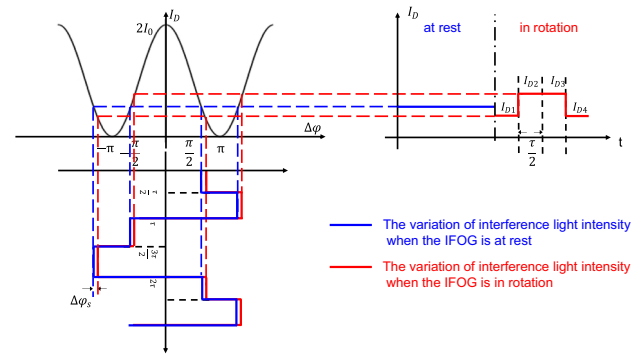


Fig. 6. Variation of I_D when the four-state modulation method is applied.

$$-1 \times I_{D1} + I_{D2} + I_{D3} + (-1) \times I_{D4} = 4I_0 \sin \theta \sin \Delta\varphi_s. \quad (11)$$

After considering the effect of the feedback link as well as the 2π reset error and demodulating it synchronously with the sequence $\{1, -1, 1, -1\}$ (each state lasts $\tau/2$), we can get,

$$I_{D1} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(\theta - \Delta\varphi_s(k - 1))]\}, \quad (12)$$

$$I_{D2} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(2\pi - \theta - \Delta\varphi_s(k - 1))]\}, \quad (13)$$

$$I_{D3} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(-\theta - \Delta\varphi_s(k - 1))]\}, \quad (14)$$

$$I_{D4} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(\theta - 2\pi - \Delta\varphi_s(k - 1))]\}, \quad (15)$$

$$I_{D1} + I_{D3} - (I_{D2} + I_{D4}) \approx -4\pi I_0 \varepsilon \sin \theta, \quad (16)$$

where $1 + \varepsilon$ represents the modulation phase magnification factor resulting from changes in environmental factors, particularly temperature.

It can be seen from Eq. (16) that the four-state modulation can correct the 2π reset error in time in every demodulation period, regardless of whether a ramp reset occurs or not. It improves the IFOG's stability against ambient temperature fluctuations at low speeds.

However, four-state modulation still cannot solve the problem of crosstalk. Theoretically, only the photodetector output voltage V_D induced by the interference light intensity I_D participates in the demodulation process. In practice, a part of the modulation voltage also participates in the demodulation process [shown in Eq. (17)] because the modulation voltage is at the same frequency as V_D ,

$$(V_D + k_{\text{noise}} V_{\text{mod}}) V_{\text{demod}} = V_D V_{\text{demod}} + k_{\text{noise}} V_{\text{mod}} V_{\text{demod}}, \quad (17)$$

where k_{noise} is the coupling coefficient, V_{demod} is the demodulation sequence, and V_{mod} is the modulation voltage shown in Fig. 5(b). Apparently, from Eq. (17), when k_{noise} is fixed, the magnitude of crosstalk can be reflected by the correlation between the modulation voltage and the demodulation sequence,

$$\begin{aligned} r &= V_{\text{mod}} V_{\text{demod}} \\ &= V_{2\pi-\theta} - V_{\theta}. \end{aligned} \quad (18)$$

Only when the modulation depth $\theta = \pi$, is the correlation coefficient r equal to 0. In this case, according to Eq. (11) the IFOG's sensitivity $dI_D/d\Delta\varphi_s$ is 0, which makes the IFOG malfunction. In other words, the four-state modulation method cannot avoid electrical crosstalk. This is a serious impediment to the pursuit of high-precision IFOGs.

3. MODULATION METHOD

Physically blocking electrical crosstalk has limited effects. To minimize the effect of crosstalk on the demodulation result,

we usually utilize modulation techniques to reduce the correlation between the modulation voltage and the demodulation sequence. Our proposed six-state modulation distinguishes itself from many other methods by its capability to eliminate the influence of crosstalk with the fixed phase delay, regardless of the modulation depth. Furthermore, it maintains simplicity and high efficiency.

From the above discussion, it is clear that the $\Delta\varphi_b$ depends on the modulation voltage difference between moments t and $t + \tau$, rather than the voltage value at a given time. For example, in Section 2.C, the modulation voltages are V_1 , V_2 , $V_1 - V_{\theta}$, and $V_2 - V_{2\pi-\theta}$ (each state lasts for $\tau/2$), and accordingly, the bias phase difference $\Delta\varphi_b = \theta$, $2\pi - \theta$, $-\theta$, and $\theta - 2\pi$. Therefore, the values of V_1 and V_2 have relatively high degrees of freedom. Based on this idea, we present a six-state modulation method that controls the modulation voltage to reduce the correlation. It removes the effect of crosstalk with the fixed phase delay at any modulation depth.

The six-state modulation method applies bias phase difference $\Delta\varphi_b$ according to Fig. 7(a), which is different from those in Zhang *et al.*'s [11], and each state lasts for $\tau/3$. The corresponding modulation voltage is displayed in Fig. 7(b).

Without considering the feedback link, employing the same analysis procedure as described in the previous section, the I_D can be calculated as follows and is shown in Fig. 8:

$$I_{D1} = I_0 [1 + \cos(\Delta\varphi_s + \theta)], \quad (19)$$

$$I_{D2} = I_0 [1 + \cos(\Delta\varphi_s + 2\pi - \theta)], \quad (20)$$

$$I_{D3} = I_0 [1 + \cos(\Delta\varphi_s - \theta)], \quad (21)$$

$$I_{D4} = I_0 [1 + \cos(\Delta\varphi_s - \theta)], \quad (22)$$

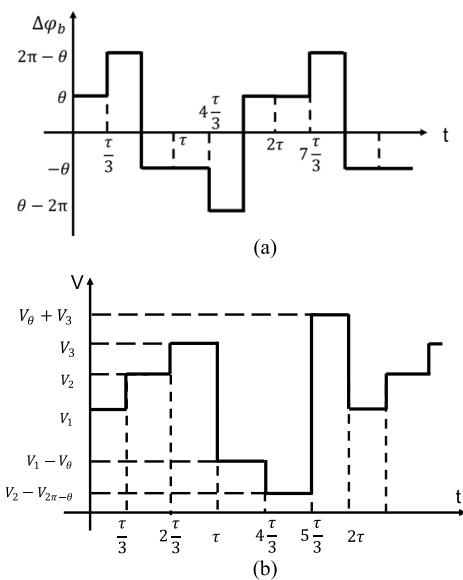


Fig. 7. (a) Phase difference $\Delta\varphi_b$ of the six-state modulation method. (b) The corresponding modulation voltage of the six-state modulation method.

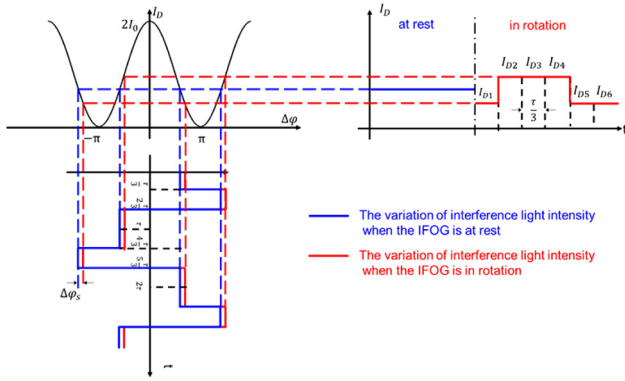


Fig. 8. Variation of I_D when the six-state modulation method is applied.

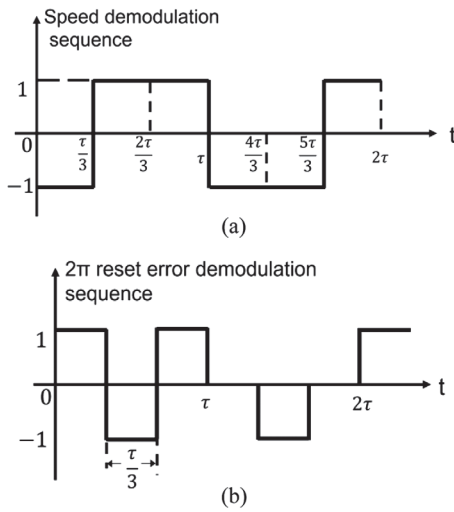


Fig. 9. (a) Speed demodulation sequence of the six-state modulation method. (b) 2π reset error demodulation sequence of the six-state modulation method.

$$I_{D5} = I_0 [1 + \cos(\Delta\varphi_s + \theta - 2\pi)], \quad (23)$$

$$I_{D6} = I_0 [1 + \cos(\Delta\varphi_s + \theta)]. \quad (24)$$

As shown in Fig. 8, when the fiber optic gyroscope is at rest, $I_D = I_0(1 + \cos \theta)$, which is a constant DC signal; when it operates at a certain rotational speed, the interference light intensity I_D manifests itself as a square-wave signal with the period of 2τ . The speed demodulation sequence and the 2π reset error demodulation sequence are depicted in Fig. 9. They are orthogonal, avoiding possible interference between the two demodulation loops. Considering the feedback loop and using the demodulation sequence in Fig. 9(a), the k th demodulation result of the six-state modulation is obtained, as shown in Eq. (25),

$$\begin{aligned} \text{demod}(k) &= I_{D2} + I_{D3} + I_{D6} - (I_{D1} + I_{D4} + I_{D5}) \\ &= 2I_0 \sin \theta \sin(\Delta\varphi_s(k) - \Delta\varphi_s(k-1)) \\ &\approx 2I_0 \sin \theta (\Delta\varphi_s(k) - \Delta\varphi_s(k-1)). \end{aligned} \quad (25)$$

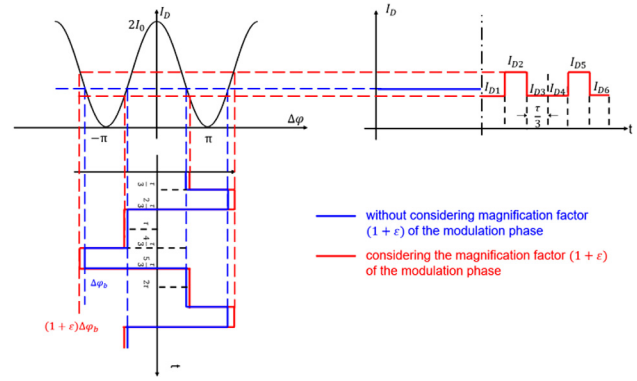


Fig. 10. Approximate variation of I_D when the six-state modulation method is applied after considering the feedback link and the 2π reset error.

Taking the feedback link and the 2π reset error into account, we get the I_D expression shown below. Considering that $\Delta\varphi_{\beta}(k) \approx -\Delta\varphi_s(k-1)$ and $1 + \varepsilon \approx 1$, the corresponding graph can be depicted in Fig. 10,

$$I_{D1} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(\theta - \Delta\varphi_s(k-1))]\}, \quad (26)$$

$$I_{D2} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(2\pi - \theta - \Delta\varphi_s(k-1))]\}, \quad (27)$$

$$I_{D3} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(-\theta - \Delta\varphi_s(k-1))]\}, \quad (28)$$

$$I_{D4} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(-\theta - \Delta\varphi_s(k-1))]\}, \quad (29)$$

$$I_{D5} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(\theta - 2\pi - \Delta\varphi_s(k-1))]\}, \quad (30)$$

$$I_{D6} = I_0 \{1 + \cos [\Delta\varphi_s(k) + (1 + \varepsilon)(\theta - \Delta\varphi_s(k-1))]\}. \quad (31)$$

Synchronously demodulating I_D with the sequence in Fig. 9(b), we can get the result

$$I_{D1} + I_{D3} - (I_{D2} + I_{D5}) \approx -4\pi I_0 \varepsilon \sin \theta. \quad (32)$$

It is obvious from Eq. (32) that the six-state modulation method retains the advantage of the four-state modulation, which can also correct the error at each demodulation cycle, enhancing resistance to temperature fluctuations at low speeds.

From Eq. (17) in Section 2.C, the influence of the crosstalk on the demodulation results can be measured by the correlation between the modulation voltage and the demodulation sequence. In practice, phase delays can be measured based on electrical crosstalk detection methods. Usually, the phase delay of each fiber optic gyroscope is concentrated around a fixed value. This paper analyzes the situation when the fixed phase delay is 0° and 60° , respectively.

The expression for the correlation between the modulation voltage with zero phase delay and the rotational speed demodulation sequence $\{-1, +1, +1, -1, -1, +1\}$ is

$$\begin{aligned}
r_1 &= -V_1 + V_2 + V_3 - (V_1 - V_\theta) - (V_2 - V_{2\pi-\theta}) + (V_3 + V_\theta) \\
&= 2V_3 - 2V_1 + V_{2\pi-\theta} + 2V_\theta \\
&= 2V_3 - 2V_1 + V_\theta + V_{2\pi}.
\end{aligned} \tag{33}$$

The expression for the correlation between the modulation voltage with zero phase delay and the 2π reset error demodulation sequence $\{+1, -1, +1, 0, -1, 0\}$ is

$$\begin{aligned}
r_2 &= V_1 - V_2 + V_3 + 0 - (V_2 - V_{2\pi-\theta}) + 0 \\
&= V_1 - 2V_2 + V_3 + V_{2\pi-\theta} \\
&= V_1 - 2V_2 + V_3 - V_\theta + V_{2\pi}.
\end{aligned} \tag{34}$$

In order to eradicate the detrimental effect of electrical crosstalk on the demodulation result, the correlation must be zero. And thus, we can get the constraint conditions on the modulation voltage, as shown in Eq. (35),

$$\begin{cases} r_1 = 0 \\ r_2 = 0 \end{cases} \Rightarrow \begin{cases} V_1 - V_3 = V_\pi + V_\theta/2 \\ V_1 - 2V_2 + V_3 = V_\theta - V_{2\pi} \end{cases}. \tag{35}$$

If the phase delay is 60° , then the expression for the correlation between the modulation voltage and the rotational speed demodulation sequence $\{-1, +1, +1, -1, -1, +1\}$ is

$$\begin{aligned}
r_1 &= -(V_3 + V_\theta) + V_1 + V_2 - V_3 - (V_1 - V_\theta) + (V_2 - V_{2\pi-\theta}) \\
&= 2V_2 - 2V_3 - V_{2\pi-\theta} \\
&= 2V_2 - 2V_3 + V_\theta - V_{2\pi}.
\end{aligned} \tag{36}$$

The expression for the correlation between the modulation voltage with 60° phase delay and the 2π reset error demodulation sequence $\{+1, -1, +1, 0, -1, 0\}$ is

$$\begin{aligned}
r_2 &= V_3 + V_\theta - V_1 + V_2 - (V_1 - V_\theta) \\
&= V_2 + V_3 - 2V_1 + 2V_\theta.
\end{aligned} \tag{37}$$

When the correlation is 0, the modulation voltage with 60° phase delay satisfies the following Eq. (38):

$$\begin{cases} r_1 = 0 \\ r_2 = 0 \end{cases} \Rightarrow \begin{cases} V_2 - V_3 = V_\pi - V_\theta/2 \\ V_2 + V_3 - 2V_1 = -2V_\theta \end{cases}. \tag{38}$$

According to Eqs. (35) and (38), as long as V_1 , V_2 , and V_3 satisfy the conditions under the corresponding phase delays, the correlation between the modulation voltage and the demodulation sequence can be zero.

Summarizing from the above discussion, apart from retaining the superiority of the four-state modulation, our six-state modulation method can also eliminate crosstalk with fixed phase delay at any modulation depth. This in turn reduces the bias phenomena and ARW, improving the accuracy of the IFOG. The six-state modulation method provides prospects for advancing IFOG development and applications.

4. EXPERIMENTS AND DISCUSSION

To examine the crosstalk reduction effects of our six-state modulation method, we conducted tests on a specific model of fiber optic gyroscope to measure the zero bias, the bias instability, and the ARW. The length of the gyroscope used is 2100 m, the diameter is about 0.14 m, and the wavelength of the light source is about 1550 nm. Our gyroscope was fixed on a marble surface at a latitude of 31.92° and a local ground speed of $-7.73^\circ/\text{h}$, employing modulation depths of $\pi/2$, $2\pi/3$, and $7\pi/8$. The testing duration was over 1 h. The output data was smoothed with a precision of one second and was analyzed. By subtracting the local ground speed, the bias of the gyroscope was obtained. What is more, the Allan deviation of the IFOG outputs was plotted to analyze the bias instability and ARW of the gyroscope.

The modulation depth has an impact on the coherence intensity noise of the light source, thereby affecting the bias instability and the ARW. However, the influence on zero bias is relatively small. Therefore, considering the space limitations, this experiment focuses only on the bias with a modulation depth of $\pi/2$. Figures 11(a) and 11(b) depict the gyroscope output versus time for four-state modulation and six-state modulation, respectively.

As seen in Fig. 11, the six-state modulation method proposed in this paper reduces the zero bias of the gyroscope by about $0.3319^\circ/\text{h}$ compared with four-state modulation. This proves its great advantage in reducing the zero bias and thus improving the output accuracy.

What is more, Fig. 12 shows Allan deviation of the IFOG outputs with modulation depths of $\pi/2$, $2\pi/3$, and $7\pi/8$ when the four-state modulation or the six-state modulation is applied. The corresponding ARWs and bias instabilities are summarized in Tables 1 and 2.

It can be seen from Table 1 that the ARW is decreased by 60%, 69.23%, and 64.29% at modulation depths of $\pi/2$, $2\pi/3$, and $7\pi/8$, respectively. Additionally, Zheng *et al.* [20] proposed a hardware method and realized an almost threefold reduction

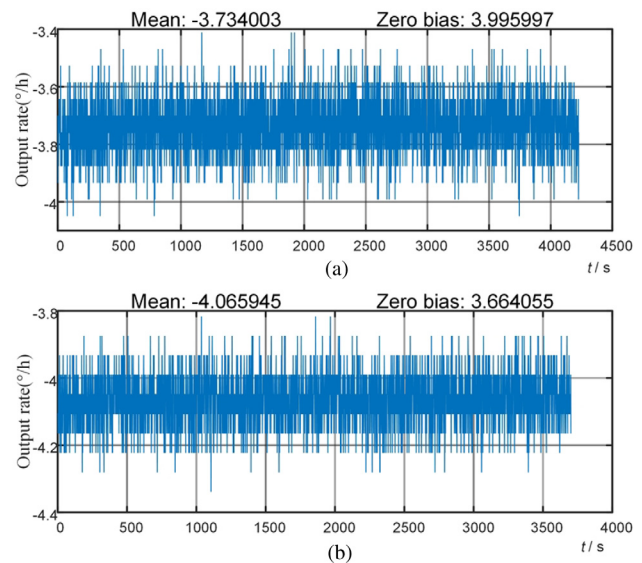


Fig. 11. (a) The gyroscope output versus time for the four-state modulation method with a modulation depth of $\pi/2$; (b) the gyroscope output versus time for the six-state modulation method with a modulation depth of $\pi/2$.

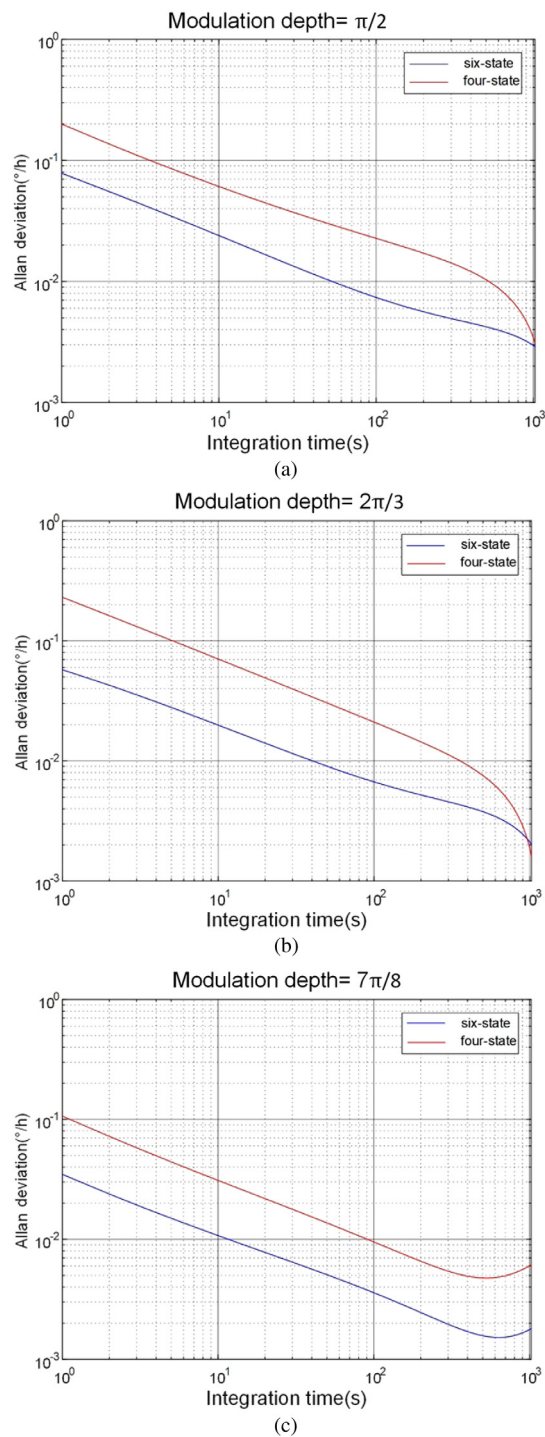


Fig. 12. Allan deviation for different modulation methods (the red and blue curves correspond to the four-state and six-state modulation methods, respectively) with modulation depths of (a) $\pi/2$, (b) $2\pi/3$, and (c) $7\pi/8$.

in the ARW. It reduces excessive relative-intensity-noise by introducing a Faraday rotator mirror, improving the ARW from 1.70 mdeg/ $\sqrt{h}$ to 0.58 mdeg/ $\sqrt{h}$ for a navigation-grade FOG. In future studies, the performance of the FOG can be further improved by introducing a variety of elimination methods.

Additionally, it can be seen from Table 2 that our method achieved a minimum bias instability of 0.0035°/h, which is

Table 1. ARWs Obtained from Fig. 12 and the Corresponding Reduction

Modulation Depth	ARW (°/ $\sqrt{h}$)		Reduction (%)
	Four-state	Six-state	
$\pi/2$	0.0035	0.0014	60.00
$2\pi/3$	0.0039	0.0012	69.23
$7\pi/8$	0.0014	0.0005	64.29

Table 2. Bias Instabilities Obtained from Fig. 12 and the Corresponding Reduction

Modulation Depth	Bias Instability (°/h)		Reduction (%)
	Four-state	Six-state	
$\pi/2$	0.0088	0.0060	31.82
$2\pi/3$	0.0068	0.0045	33.82
$7\pi/8$	0.0050	0.0035	30.00

Table 3. Bias Instability Comparison of State-of-the-Art

Year	Description	Bias Instability (°/h)
2020 [22]	A gyroscope using Brillouin ring lasers on a silicon chip	3.6°/h
2022 [23]	The proposed RFOG is driven with a broadband light source, using a fiber resonator with a finesse of 63 containing 100-m long fiber	0.009°/h
2023 [24]	A novel compensation method based on the coupling error characterization and calibration by the least square method algorithm	0.008°/h
This work	A six-state modulation method to improve the accuracy of IFOG by reducing the electrical crosstalk	0.0035°/h

superior to the state-of-the-art mentioned in [21], and the comparison is shown in Table 3.

To sum up, it can be concluded that the proposed six-state modulation method in this study achieves a reduction of over 60% in the ARW and decreases the bias instability by approximately 30% for given modulation depths. These findings demonstrate the practicality of our method and validate its effectiveness in minimizing electrical crosstalk.

5. CONCLUSION

In this paper, we propose what we believe is a novel six-state modulation method for IFOGs. By eliminating the correlation between the modulation voltage and the demodulation sequence, electrical crosstalk is suppressed, thus improving the accuracy of the fiber optic gyroscope. Moreover, in theory, this six-state modulation method can potentially eliminate crosstalk with a fixed phase delay at any modulation depth. Experimental results indicate that compared to the traditional four-state modulation method, this approach reduces the zero bias of the gyroscope by about 0.3319°/h at the modulation depth of $\pi/2$. Moreover, it reduces the bias instability by approximately 30%

and the ARW by over 60%. The study suggests that it is a highly attractive modulation method for further advancements in high-precision gyroscopes.

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Data availability. The authors confirm that the data supporting the findings of this study are available in the paper.

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